

CPP NOTES -DAY 29

Queue

A Queue Data Structure is a fundamental concept in computer science used for storing and managing data in a specific order.

- It follows the principle of "First in, First out" (FIFO), where the first element added to the queue is the first one to be removed.
- It is used as a buffer in computer systems where we have speed mismatch between two devices that communicate with each other. For example, CPU and keyboard and two devices in a network
- Queue is also used in Operating System algorithms like CPU Scheduling and Memory Management, and many standard algorithms like Breadth First Search of Graph, Level Order Traversal of a Tree.

Self Referential pointer

A self-referential pointer (often called a self-pointer) is a pointer within a structure (or class in object-oriented programming) that points to an instance of the same structure or class. It's a common concept in languages like C, C++, and sometimes used indirectly in object-oriented languages.

```
struct Node {  
    int data;  
  
    struct Node* next; // self-referential pointer  
};
```

In this case, next is a pointer to another Node structure. It allows you to chain nodes together to form linked structures.